

JINYU FANG

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EDUCATION

Boston University

Expected May 2028

B.S., Mechanical Engineering, Minor in Chemistry, Concentration in Energy Technologies & Sustainability

GPA: 3.77/4.00

- Relevant Coursework: Statistical Thermodynamics, Material Science, Instrumental Chemical Analysis, Organic Chemistry, Mechanics of Materials, Programming for Engineers

EXPERIENCE

BU College of Engineering

Boston, MA

EK301: Engineering Mechanics Grader

Sep 2026 – Dec 2026

- Grade problem sets and exams for 90+ undergraduates on a 7-day turnaround, flagging recurring misconceptions for instructors

BU Mesostructured Materials & Devices Lab

Boston, MA

Undergraduate Research Assistant

Jul 2026 – Present

- Developed and cast low-tortuosity LFP battery electrodes using nonequilibrium soft-matter processing to enhance ion transport in high-energy-density batteries
- Engineered dual-scale porosity to reduce capacity loss during fast lithiation, targeting electric vehicle (EV) battery performance
- Characterized electrode architectures via EIS and current cycling to optimize the balance between energy and power density
- Fabricated 3D carbon sheet architectures for applications ranging from carbon capture to high-energy-density batteries

University of Sydney

Sydney, Australia

Sydney Sustainability Program Volunteer

Feb 2026 – Jun 2026

- Selected for a competitive sustainability leadership program; completed 4 of the UN Sustainable Development Goals (SDGs)
- Volunteered 12+ hours with Pocket City Farms and BRAD Banish Recycling; studied the Michael Mobbs off-grid house

Sustainable House

Sydney, Australia

Sustainability Analyst Intern

Jan 2026 – Jun 2026

- Analyzed carbon, climate, and food-waste data (Excel) for outreach graphs and suburb-wide waste-elimination goals
- Built and ran aerobic composting units ("Coolseats"), producing 5 tonnes of compost; tutored primary school students
- Authored technical articles and reports turning complex sustainability data into actionable insights for a global audience

BU College of Engineering

Boston, MA

EK301: Engineering Mechanics Learning Assistant

Sep 2025 – Dec 2025

- Facilitated peer learning in statics; presented a post-lecture feedback poster to 10+ peers, professors, and professionals

PROJECTS

BU Mars Rover Club, Life Detection Member

Oct 2024 – Present

- Researched total organic carbon (TOC) processes and designed organic carbon procedures for detecting life on the rover
- Tested pipette, agitation, and reaction configurations for Organisense to analyze organic carbon in soil
- Developed a DNS assay standard operating procedure (SOP) for determining organic material in soil
- Created a cuvette holder with a rack and pinion for testing organic materials

Color Sorter, Engineering Design Student

Jun 2026 – Aug 2026

- Engineered a compact (2 ft × 2 ft) automated optical sorter that classifies and routes multi-colored objects across 5 color channels using color-sensing algorithms on an embedded microcontroller
- Designed a hands-free feeding and transparent multi-bin sorting system for clean, continuous batch runs
- Optimized actuation and sensing to reliably sort 21 pieces per minute; managed parts and fabrication within a \$200 budget

Steel C-Channel Beam Deflection Analysis, Structural Analyst

May 2026

- Loaded a simply supported steel C-channel off-center in 44.5 N steps to 222.5 N, measuring deflection with dial gauges
- Derived closed-form deflection and bending-moment expressions, predicting every deflection within 10% of measurement
- Showed a linear load-deflection response ($R^2 \geq 0.998$) with full elastic recovery

LEADERSHIP/ACTIVITIES

Boston University Allocations Board, Secretary

Dec 2024 – Present

- Distribute \$1,000,000 in need-based funding to 450 undergraduate organizations; manage 40+ emails weekly within 48 hours

SKILLS

Sustainability: Energy Auditing, Life-Cycle & Carbon Emissions Analysis, Waste Optimization, UN SDGs 11, 13, 15, 17

Software: Python, MATLAB, Excel, SolidWorks, Autodesk Fusion 360, Arduino, LaTeX, Microsoft Office

Instrumentation: Electrochemical Impedance Spectroscopy (EIS), FTIR, Raman, HPLC, GC-MS, UV-Vis, ^1H NMR, XRD

Laboratory/Fabrication: Electrode Casting, 3D Printing, Laser Cutting, Soldering, Circuit Design, Quantitative Analysis